MTH 102: Probability and Statistics

Midsem Exam

16/02/2019

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Explain your answers. Show your steps. No gadgets allowed. Approximate calculations are fine. Good luck!

Question 1. 10 **marks** A number is drawn from a Gaussian distribution with mean μ and standard deviation σ. A priori your belief is therefore summarized by the PDF $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$, $x \in (\infty, \infty)$. You are told that the drawn number was larger than the mean. Express your revised belief given the information.

Suppose two such numbers are drawn independently of each other. In the absence of any information regarding the outcomes of the draw, calculate the probability that both the numbers were either larger than the mean or smaller than the mean?

Question 2. 20 **marks** You have submitted your application to the Ministry of Procrastination (MoP). Historically, the time X MoP takes to respond is well described by an exponential distribution with mean 1 year. We have $f_X(x) = \mu e^{-\mu x}, x \geq 0$, with mean $1/\mu = 1$. Calculate the CCDF. Suppose that MoP has not responded to your application for 2 years. Write down this event in terms of the random variable X. Given this information, calculate the conditional CCDF of X and the corresponding PDF. Comment on how the PDF compares with $f_X(x)$.

Question 3. 35 **marks** A game of TT has two players that occupy opposite ends of a table. A TT game may be thought of as multiple rallies between the players. A rally is a sequence of alternate hits by the players that ends when a player's hit ends in error. We will call an erroneous hit a *miss*. As an example, a rally may consist of a *hit* by player 1, followed by a hit by player 2, and so on, and may end with a miss by player 2. Suppose that player 1 hits with probability 0.6 and misses with probability 0.4. Player 2 on the other hand hits with probability 0.4. Answer the following questions. Assume a rally always starts with player 1.

1) Calculate the probability that a rally has less than three hits?
2) Calculate the probability that a rally ends with a miss by player 1. If you will, leave your answer as a fraction.
3) Calculate the expectation of the length of a rally, where the length includes all hits and the final miss.
4) Calculate the expectation of the number of rallies, amongst 10 rallies, that end due to a miss by player 2?

Question 4. 35 **marks** You start your day at a bus stop excited to reach the probability and stats lecture on time. A bus arrives a random B minutes later where B is exponentially distributed with mean $1/\mu = 10$ minutes. The bus takes 45 minutes to reach the lecture's venue. If the bus doesn't arrive for 10 minutes, you decide to take an Uber. The Uber arrives a random U minutes later where U is exponentially distributed with mean 5 minutes. The Uber takes 30 minutes to reach the lecture's venue. If the Uber doesn't arrive within 5 minutes, you decide to take an auto. The auto arrives A minutes later where A is exponentially distributed with mean 5 minutes. The auto, like Uber, takes 30 minutes to reach the lecture's venue. If the auto doesn't arrive within 5 minutes you decide not to attend the lecture and head home, which is a 5 minute walk from the bus stop. Answer the following questions.

1) Over any selection of 10 days, what is the probability that you take the Uber 2 days, the bus for 4 days, and the auto for 2 days. Choices are independent of each other. Don't simplify exponentials, if any.
2) Calculate the expected value of the time you spend waiting for transport.
3) Assuming you reach the lecture, calculate the expected value of the total time it takes from the start of day.
4) Calculate the expected value of the total time spent by you from the start of your day till you reach the lecture venue or are back home.

Question 1. 10 **marks** A number is drawn from a Gaussian distribution with mean μ and standard deviation σ. A priori your belief is therefore summarized by the PDF $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$, $x \in (-\infty, \infty)$. You are told that the drawn number was larger than the mean. Express your revised belief given the information.

Suppose two such numbers are drawn independently of each other. In the absence of any information regarding the outcomes of the draw, calculate the probability that both the numbers were either larger than the mean or smaller than the mean?

The event that the number drawn is larger than the mean is $\{X > \mu\}$

Our revised belief is $f_{X|\{X>\mu\}}(x)$ calculated for all $x \in (-\infty, \infty)$.

$$f_{X|\{X>\mu\}}(x) = \begin{cases} \dfrac{f_X(x)}{P[X>\mu]} & x > \mu \\ 0 & \text{otherwise.} \end{cases}$$

where $P[X>\mu] = 0.5$. [This is simply because a Gaussian PDF is symmetric around μ].

Four mutually exclusive outcomes

0 0
0 1
1 0
1 1

where $0 \Rightarrow$ the drawn number is $\le \mu$

$1 \Rightarrow$ the number is $> \mu$.

The event of interest is $00 \cup 11$

$\therefore P[00 \cup 11] = P[00] + P[11]$

Since the outcomes of the draws are independent we can write:

$$P[00 \cup 11] = P[0]\,P[0] + P[1]\,P[1]$$

$$= \left(\tfrac{1}{2}\right)\left(\tfrac{1}{2}\right) + \left(\tfrac{1}{2}\right)\left(\tfrac{1}{2}\right)$$

$$= 1/2.$$

Question 2. 20 marks You have submitted your application to the Ministry of Procrastination (MoP). Historically, the time X MoP takes to respond is well described by an exponential distribution with mean 1 year. We have $f_X(x) = \mu e^{-\mu x}, x \geq 0$, with mean $1/\mu = 1$. Calculate the CCDF. Suppose that MoP has not responded to your application for 2 years. Write down this event in terms of the random variable X. Given this information, calculate the conditional CCDF of X and the corresponding PDF. Comment on how the PDF compares with $f_X(x)$.

Given $f_X(x) = \mu e^{-\mu x}, \; x \geq 0.$

CCDF calculates the probability $P[X > x]$ for any $x \in \mathbb{R}$.

$$P[X>x] = \begin{cases} \int_x^\infty f_X(x)\,dx = e^{-\mu x} & \forall x \geq 0. \\ 1 & \forall x < 0. \end{cases}$$

From the definition of the PDF. It is non-zero only for $x > 0$.

The event that is known to have occurred is $\{X > 2\}$

We want $P[X > x \mid X > 2]$ for all x.

Consider $x \leq 2$:

$$P[X > x \mid X > 2] = 1.$$

For $x > 2$:

$$P[X > x \mid X > 2] = \frac{P[X > x, X > 2]}{P[X > 2]}$$

$$= \frac{P[X > x]}{P[X > 2]} \qquad x > 2.$$

$$= \frac{e^{-\mu x}}{e^{-\mu 2}} = e^{-\mu(x-2)} \qquad x > 2$$

To summarize:

$$P[X > x \mid X > 2] = \begin{cases} e^{-\mu(x-2)} & x > 2 \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = X - 2$. This is the additional time you will have to wait past the two years you have already spent waiting.

$$P[Y > y \mid X > 2] = \begin{cases} e^{-\mu y} & y > 0 \\ 0 & \text{otherwise.} \end{cases}$$ PDF ?? (Easy to do)

Additional Commentary; HW at end.

Turns out that even after having waited for two hours the CCDF of the additional time you will wait is that of an exponential RV with rate μ. This is exactly the same RV at the beginning of your wait. Therefore, given that the wait time is exponentially distributed, even after having waited for time z, the remaining wait time is still distributed identically to that at the beginning. This property is unique to the Exponential & Geometric distributions. It is called the "memoryless property". In the above example, the length of time you have waited was inconsequential to the additional time you will have to wait.

HW: Try calculating the above assuming that the wait time $f_X(x)$ is $\text{Unif}(a, b)$, $b > a \geq 0$. Does the Uniform distribution have the memoryless property?

Also, do the above for when the wait is a geometric RV.

Question 3. **35 marks** A game of TT has two players that occupy opposite ends of a table. A TT game may be thought of as multiple rallies between the players. A rally is a sequence of alternate hits by the players that ends when a player's hit ends in error. We will call an erroneous hit a *miss*. As an example, a rally may consist of a *hit* by player 1, followed by a hit by player 2, and so on, and may end with a miss by player 2. Suppose that player 1 hits with probability 0.6 and misses with probability 0.4. Player 2 on the other hand hits with probability 0.4. Answer the following questions. Assume a rally always starts with player 1.

1) Calculate the probability that a rally has less than three hits? 5
2) Calculate the probability that a rally ends with a miss by player 1. If you will, leave your answer as a fraction. 10
3) Calculate the expectation of the length of a rally, where the length includes all hits and the final miss. 10
4) Calculate the expectation of the number of rallies, amongst 10 rallies, that end due to a miss by player 2? 10

(1) A rally has less than 3 hits if it has either 0, 1, or 2 hits.

0 hits ⇒ Player 1 misses at the start
1 hit ⇒ Player 1 hits & Player 2 misses
2 hits ⇒ Player 1 hits, Player 2 hits, Player 1 misses

The above three events are mutually exclusive.

$$P[\text{Rally has} < 3 \text{ hits}] = P[0 \text{ hits}] + P[1 \text{ hit}] + P[2 \text{ hits}]$$

$$= (0.4) + (0.6)(0.6) + (0.6)(0.4)(0.4)$$
$$= (0.4) + (0.6)^2 + (0.6)(0.4)^2$$
$$= (0.4) + (0.6)[(0.6) + 0.16]$$
$$= 0.4 + (0.6)(0.76)$$
$$= 0.856.$$

(2) $P[\text{Rally ends with a miss by player 1}] = ?$

Let M_i be a miss by player i.
Let H_i be a hit by player i. $\} \; i \in \{1, 2\}$.

$$P[\text{Rally ends with a miss by player 1}]$$
$$= P[M_1 \cup (H_1 H_2 M_1) \cup (H_1 H_2 H_1 H_2 M_1) \cup \cdots]$$
$$= P[M_1] + P[H_1 H_2 M_1] + P[H_1 H_2 H_1 H_2 M_1] + \cdots$$
$$= (0.4) + (0.6)(0.4)(0.4) + (0.6)(0.4)(0.6)(0.4)(0.4) + \cdots$$
$$= (0.4)\left[1 + (0.6)(0.4) + ((0.6)(0.4))^2 + \cdots\right]$$
$$= \frac{0.4}{1 - 0.24} = \frac{0.4}{0.76}.$$

(3) Let RV L be the length of a rally.
Clearly, $S_L \in \{1, 2, \cdots\}$.
L is a discrete RV with a countably infinite range space.

The event $\{L = 1\}$ is the same as the event M_1.

$\{L=2\} \iff \{H_1 M_2\}$
$\{L=3\} \iff \{H_1 H_2 M_1\}$
$\{L=4\} \iff \{H_1 H_2 H_1 M_2\}$
⋮

$P[L=1] = P[M_1] = 0.4$
$P[L=2] = P[H_1 M_2] = P[H_1]P[M_2] = (0.6)(0.6)$
$P[L=3] = P[H_1 H_2 M_1] = (0.6)(0.4)(0.4)$
$P[L=4] = P[H_1 H_2 H_1 M_2] = (0.6)(0.4)(0.6)(0.6)$
⋮

For $k \in \{1, 2, \cdots\}$

$P[L = 2k] = P[k$ hits by player 1 and $(k-1)$ hits by player 2 and rally ends with a miss by player 2$]$

$$= (0.6)^k (0.4)^{k-1} (0.6)$$

$$P[L = 2k-1] = (0.6)^{k-1} (0.4)^{k-1} (0.4)$$

$$E[L] = \sum_{k=1}^{\infty} 2k (0.6)^k (0.4)^{k-1} (0.6) + \sum_{k=1}^{\infty} (2k-1)(0.6)^{k-1}(0.4)^{k-1}(0.4)$$

8/10 if E[L] is correctly stated but not simplified

series Calculation

$$= 2(0.6)^2 \sum_{k=1}^{\infty} k (0.24)^{k-1} + 2(0.4) \sum_{k=1}^{\infty} k (0.24)^{k-1} - \sum_{k=1}^{\infty} (0.4)(0.24)^{k-1}$$

2/10

$$\sum_{k=1}^{\infty} k x^{k-1} = \frac{d}{dx} \sum_{k=0}^{\infty} x^k = \frac{d}{dx}\left(\frac{1}{1-x}\right) = \left(\frac{1}{1-x}\right)^2$$

$$\therefore E[L] = \frac{2((0.6)^2 + 0.4)}{(0.76)^2} - \frac{0.4}{0.76} \approx \underline{\underline{2.1}}$$

(4) It is useful to think of each of the 10 rallies to be independent Bernoulli trials. Each trial has two outcomes: (T) A rally that ends with a player 1's miss and (H) A rally that ends with a player 2's miss.

The number of rallies amongst 10 that end with a player 2's miss is a Binomial RV with parameters $(10, p)$, where

$$p = P[H]$$
$$= \sum_{k=1}^{\infty} (0.6)^k (0.4)^{k-1} (0.6)$$

[This we calculated in part (3) of the question]

$$= (0.6)^2 \sum_{k=1}^{\infty} (0.24)^{k-1}$$
$$= \frac{(0.6)^2}{0.76}$$

You could also calculate this using the result in (2), in which you essentially calculated $P[T]$.

From part (2) we know that $P[T] = \frac{0.4}{0.76}$

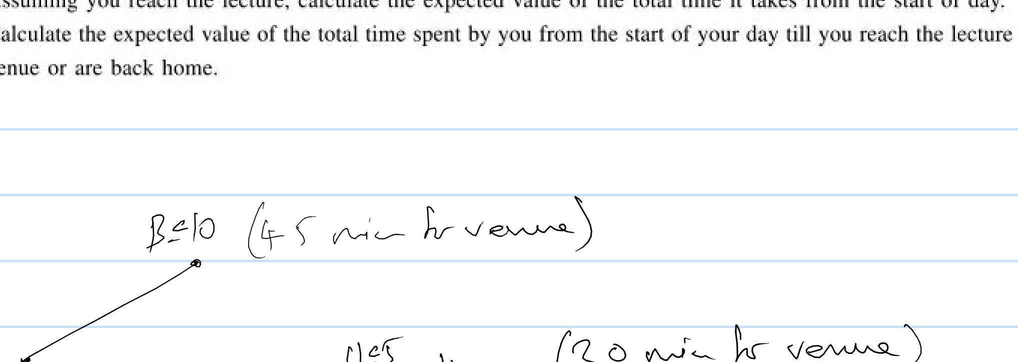
$$\therefore P[H] = \frac{0.36}{0.76}.$$

The average of interest $= (10)\left(\frac{0.36}{0.76}\right)$

$$= \frac{360}{76} \approx 4.5.$$

Question 4. 35 **marks** You start your day at a bus stop excited to reach the probability and stats lecture on time. A bus arrives a random B minutes later where B is exponentially distributed with mean $1/\mu = 10$ minutes. The bus takes 45 minutes to reach the lecture's venue. If the bus doesn't arrive for 10 minutes, you decide to take an Uber. The Uber arrives a random U minutes later where U is exponentially distributed with mean 5 minutes. The Uber takes 30 minutes to reach the lecture's venue. If the Uber doesn't arrive within 5 minutes, you decide to take an auto. The auto arrives A minutes later where A is exponentially distributed with mean 5 minutes. The auto, like Uber, takes 30 minutes to reach the lecture's venue. If the auto doesn't arrive within 5 minutes you decide not to attend the lecture and head home, which is a 5 minute walk from the bus stop. Answer the following questions.

1) Over any selection of 10 days, what is the probability that you take the Uber 2 days, the bus for 4 days, and the auto for 2 days. Choices are independent of each other. Don't simplify exponentials, if any.
2) Calculate the expected value of the time you spend waiting for transport.
3) Assuming you reach the lecture, calculate the expected value of the total time it takes from the start of day.
4) Calculate the expected value of the total time spent by you from the start of your day till you reach the lecture venue or are back home.



$P[\text{You take Uber}] = P[B>10, U\le 5]$
$= P[B>10]\,P[U\le 5] \triangleq P_U$

$P[\text{You take bus}] = P[B\le 10] \triangleq P_B$

$P[\text{You take auto}] = P[B>10, U>5, A\le 5]$
$= P[B>10]\,P[U>5]\,P[A\le 5] \triangleq P_A$

Note that we assumed the RV(s) B, U, A are mutually independent. This is a reasonable assumption.

Let P_H be the probability that you head home.

$P_H = 1 - P_U - P_B - P_A$

$P_B = P[B\le 10] = 1 - e^{-\left(\frac{1}{10}\right)10} = 1 - e^{-1}$

$P_U = e^{-1}(1-e^{-1})$

$P_A = e^{-1}e^{-1}(1-e^{-1}) = e^{-2}(1-e^{-1})$

(1) The probability of interest is:

$${}^{10}C_4\, P_B^4\; {}^{6}C_2\, P_U^2\; {}^{4}C_2\, P_A^2\; {}^{2}C_2\, P_H^2$$

$$= {}^{10}C_4\, {}^{6}C_2\, {}^{4}C_2\, {}^{2}C_2\, P_B^4 P_U^2 P_A^2 P_H^2$$

$$= \frac{10!}{6!\,4!}\,\frac{6!}{4!\,2!}\,\frac{4!}{2!\,2!}\,(1)\, P_B^4 P_U^2 P_A^2 P_H^2$$

$$= \frac{10!}{4!\,2!\,2!\,2!} P_B^4 P_U^2 P_A^2 P_H^2$$

(2) Let W be the time you wait.

$$E[W] = E[W \mid B\le 10]\,P[B\le 10] + E[W\mid B>10, U\le 5]\,P[B>10, U\le 5] + E[W\mid B>10, U>5, A\le 5]\,P[B>10, U>5, A\le 5] + E[W\mid B>10, U>5, A>5]\,P[B>10, U>5, A>5]$$

Note that we used the fact that we have an event space that consisting the events: You take the bus, You take UBER, You take an Auto, You go home.

Further note that

$E[W\mid B>10, U\le 5] = E[10+U \mid B>10, U\le 5]$
$= E[10+U\mid U\le 5]$ $\{\because B$ and U are independent
$= 10 + E[U\mid U\le 5]$

$E[W\mid B>10, U>5, A\le 5] = 15 + E[A\mid A\le 5]$

$E[W\mid B>10, U>5, A>5] = 10+5+5 = 20.$

B, U, A are exponential RV(s).

Consider $X \sim Exp(\mu)$

$$E[X\mid X\le a] = \int_0^a x\, f_{X\mid X\le a}(x)\,dx$$

$$f_{X\mid X\le a}(x) = \begin{cases} \dfrac{f_X(x)}{P[X\le a]} & x\le a \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore E[X\mid X\le a] = \int_0^a x\,\frac{\mu e^{-\mu x}}{1-e^{-\mu a}}\,dx$$

$$= \frac{-1}{1-e^{-\mu a}}\int_0^a x\, de^{-\mu x}$$

$$= \frac{-1}{1-e^{-\mu a}}\left\{\left(xe^{-\mu x}\right)_0^a - \int_0^a e^{-\mu x}dx\right\}$$

$$= \frac{-1}{1-e^{-\mu a}}\left\{ae^{-\mu a} + \left(\frac{1}{\mu}(e^{-\mu a}-1)\right)\right\}$$

$$= \frac{1}{\mu} - \frac{a e^{-\mu a}}{1-e^{-\mu a}}$$

[It is instructive to note that this goes to $1/\mu$ in the limit as $a\to\infty$. Does this make sense? If so, why?]

At $a = 1/\mu$, $E\left[X \mid X\le \frac{1}{\mu}\right] = \frac{1}{\mu} - \left(\frac{e^{-1}}{1-e^{-1}}\right)\frac{1}{\mu}$

$$= \frac{1}{\mu}\left[\frac{1-2e^{-1}}{1-e^{-1}}\right]$$

$\therefore E[W\mid B\le 10] = 10\left(\frac{1-2e^{-1}}{1-e^{-1}}\right)$

$E[W\mid B>10, U\le 5] = 5\left(\frac{1-2e^{-1}}{1-e^{-1}}\right) + 10$

$E[W\mid B>10, U>5, A\le 5] = 5\left(\frac{1-2e^{-1}}{1-e^{-1}}\right) + 15$

$E[W\mid B>10, U>5, A>5] = 20.$

$P[B\le 10] = P_B$

$P[B>10, U\le 5] = P_U$ and so on...

Thus, we can now calculate $E[W]$.

(3) We now want

$E[\text{Total Time} \mid \text{You reach the lecture}]$

Where Total time = Wait time W + Travel time

$E[\text{Total time}\mid \text{You reach the lecture}]$
$= E[W\mid \text{You reach the lecture}]$
$+ E[\text{Travel time}\mid \text{You reach the lecture}]$

Let R be the event that you reach the lecture. Note that given that R has taken place the events $\{B\le 10\}$, $\{B>10, U\le 5\}$, $\{B>10, U>5, A\le 5\}$ form an event space. Therefore:

$$E[W\mid R] = E[W\mid B\le 10, R]\,P[B\le 10\mid R] + E[W\mid B>10, U\le 5, R]\,P[B>10, U\le 5\mid R] + E[W\mid B>10, U>5, A\le 5, R]\,P[B>10, U>5, A\le 5\mid R]$$

Note that the events

$\{B\le 10, R\} \Leftrightarrow \{B\le 10\}$

$\{B>10, U\le 5, R\} \Leftrightarrow \{B>10, U\le 5\}$

$\{B>10, U>5, A\le 5, R\} \Leftrightarrow \{B>10, U>5, A\le 5\}$

So each of the conditional expectations are as calculated before.

As regards the probabilities:

$$P[B\le 10\mid R] = \frac{P[B\le 10, R]}{P[R]} = \frac{P[B\le 10]}{P[R]}$$

$$= \frac{P[B\le 10]}{(1-P_H)}$$

Others can be calculated similarly. In summary, the probabilities should be calculated by dividing the corresponding probabilities in part (2) by $P[R]$.

We can calculate $E[\text{Travel time}\mid \text{You reach the lecture}]$ as

$(45)\,P[\text{You took the bus}\mid R]$
$+ 30\,P[\text{You took Uber}\mid R]$
$+ 30\,P[\text{You took Auto}\mid R]$

$$= 45\frac{P_B}{P[R]} + 30\frac{P_U}{P[R]} + \frac{30 P_A}{P[R]}.$$

(d) Ideally part (d) must follow part (b).

In addition to part (b) we need

$E[\text{Travel Time}]$
$= 45\,P[\text{You took the bus}]$
$+ 30\,P[\text{You took Uber}]$
$+ 30\,P[\text{You took the Auto}]$
$+ 5\,P[\text{You went home}]$